Fraud Detection Modeling and Analysis

MATH-509 Final Project

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Why It Matters?

- Usually the number of frauds that happen especially in financial institutions, like banks, are very low.
- However, even a single significant fraud may lead to a huge deficit and potential loss.
- These fraud cases have a major potential to impact not only the victims, but also the country’s economy itself.

Real World Scenario

- A normal fraud detector will yield a 99.9% accuracy if it just says “no fraud” due to an extremely high class imbalance.

- Therefore, a normal model may not be able to capture uncertainty associated with these rare events.

- However, due to the potential risk of every fraud being highly damaging, it becomes necessary to predict it in a reliable way.

Project Objectives

1. **Probability, Prediction and Threshold of frauds:**
   We find the probability if a transaction is fraud or not and the optimal threshold beyond which a transaction is flagged as fraud.

2. **Least amount of money lost:**
   For example: An incident of having $1 fraud with 80% probability is less important than having $10,000 fraud with 30% probability. So we prioritize the latter to save money.

3. **Range of Frauds:**
   The percentage of frauds getting detected given a dataset should be balanced and reflect realistic scenario.

Model and Objectives

The Model Used

Objective 1

We used MCMC model to predict the probability of a given transaction being fraud.

Objective 2

We applied gradient descent to minimize the expected loss in order to apply weights to our probabilities.

Objective 3

Predicted a range which will give us an optimal real life percent fraud detection from the given dataset.

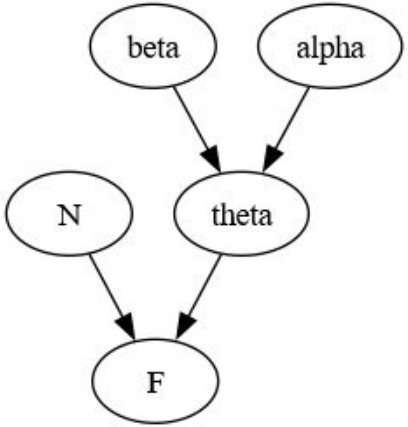
Model and Work: Objective 1

1. We perform stratified split on the dataset into 80% training and 20% testing.

2. Then we performed Logistic Regression, Random forest and SVM for a baseline comparison.

3. Now, we split the training data into 25% fraud and 75% normal and perform a 4-chain MCMC sampling on it. (Data: 240,000 -> 1800 transactions approx.).

4. Lastly, we check on all thresholds and test the model using 20% test data for accuracy comparison on baseline.



Model and Work: Objective 2

1. For this objective, we need to minimize the Expected Loss.

2. We assume that for every False Positive transaction (the transaction was flagged as fraud when it wasn't and now the consumer time and company resources are wasted to unblock the credit card) we lose $10.

3. However, we lose the entire amount in case of False Negative (where a transaction was fraud which we fail to catch and the consumer now has to be compensated).

4. We achieve this by adding a dynamic threshold, we use gradient descent algorithmically to minimize the expected loss and find a stopping point beyond which the threshold will stop decreasing (otherwise our model will constantly flag high amount transactions).

Model and Work: Objective 3

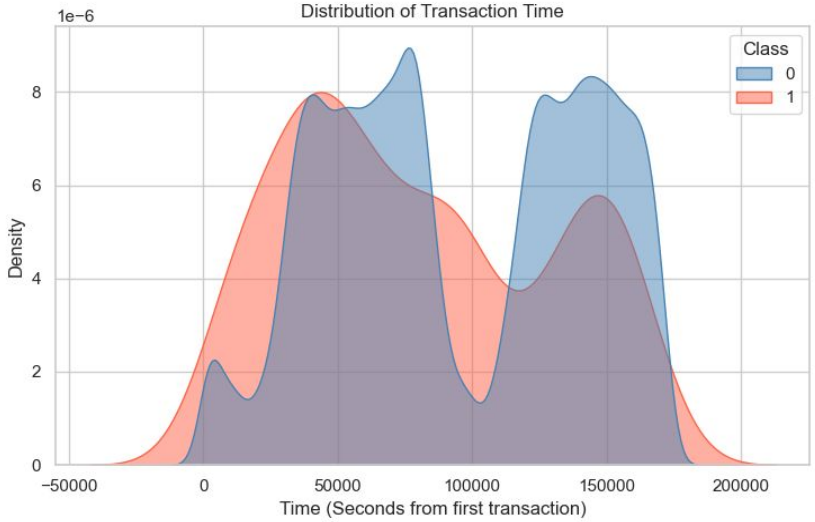
1. Finally, we check the daily transaction density and find the Confidence Interval for Fraudulent Transactions.

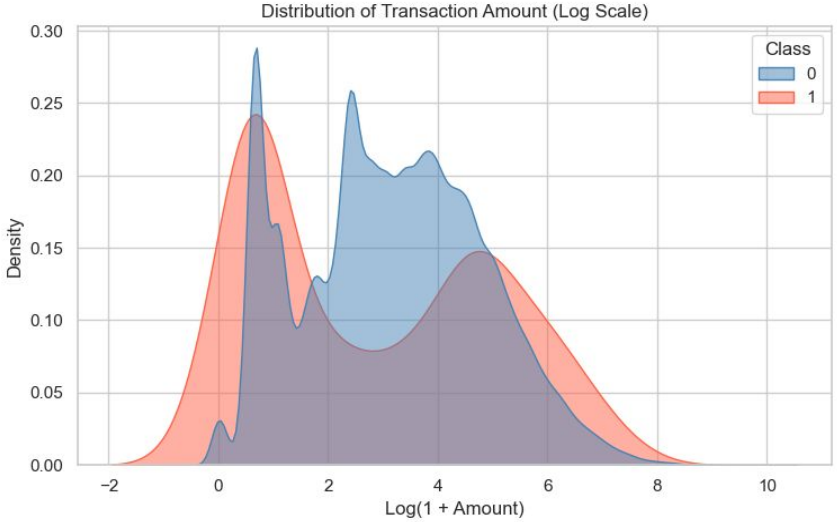
2. If the amount of Fraudulent Transactions exceed beyond this interval, the number for frauds have exceeded the normal limit and is a cause for concern.

3. If the amount of Fraudulent Transactions are below the interval, our system has disproportionately less number of frauds than normal which can also be problematic, as this would mean that many consumers are not aware of a fraud (a criteria by which we determine a fraud has occurred is when a customer reports such).

Analysis, Limitations and Future Work

Dataset Analysis





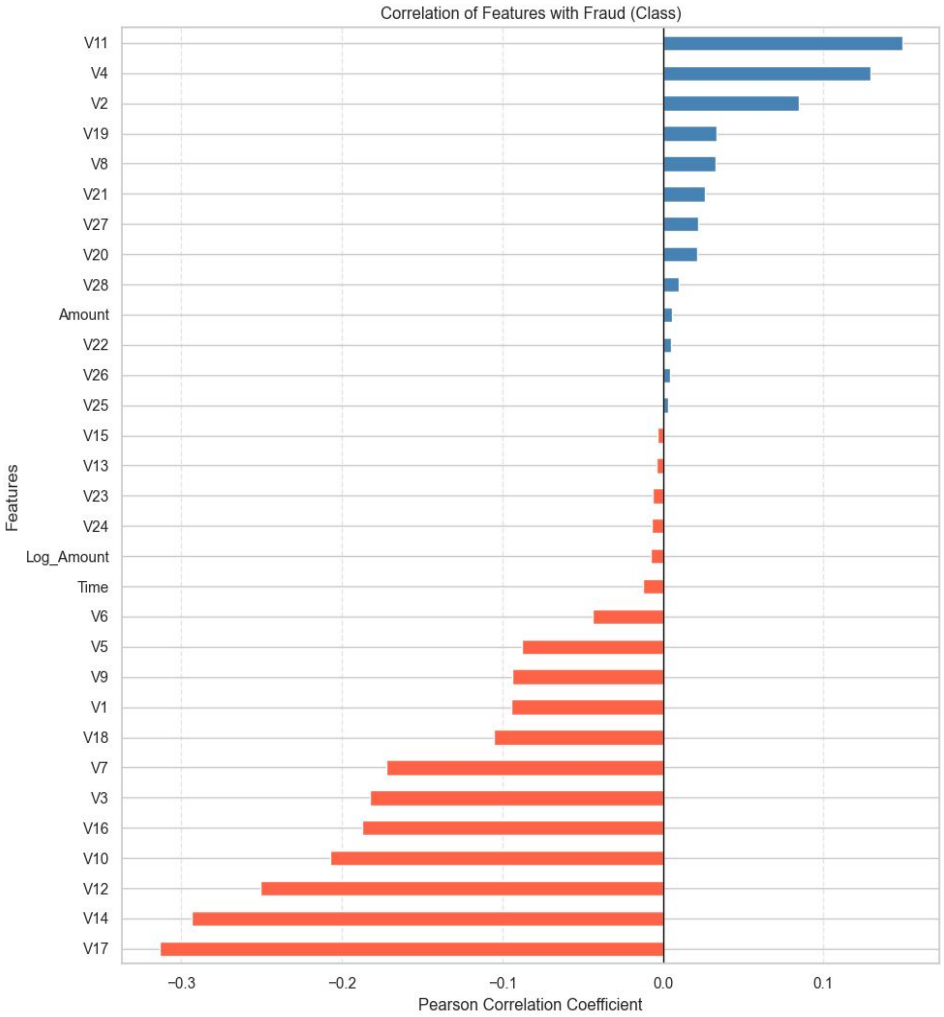
Class 0 (blue) represents a normal transaction.
Class 1 (red) represents a fraudulent transaction.

Variable Correlation

Observations:

V4, V11 are the two highly positively correlated variables.

V12, V14, V17 are the three highly negatively related variables.

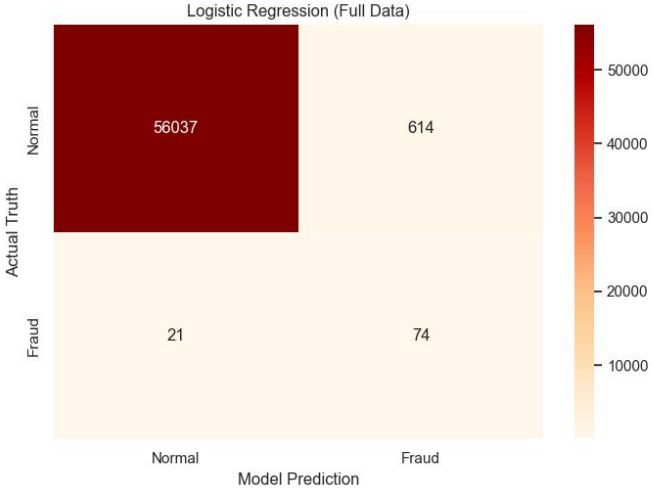


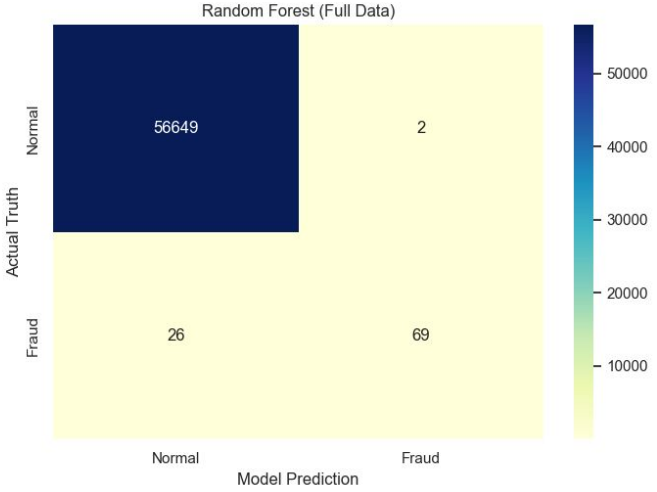
Baseline

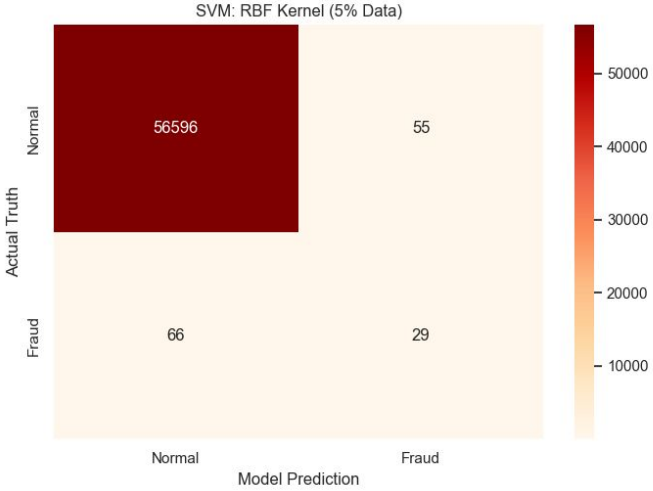
Training Set: 226,980 rows
Normal: 226,602
Fraud: 378

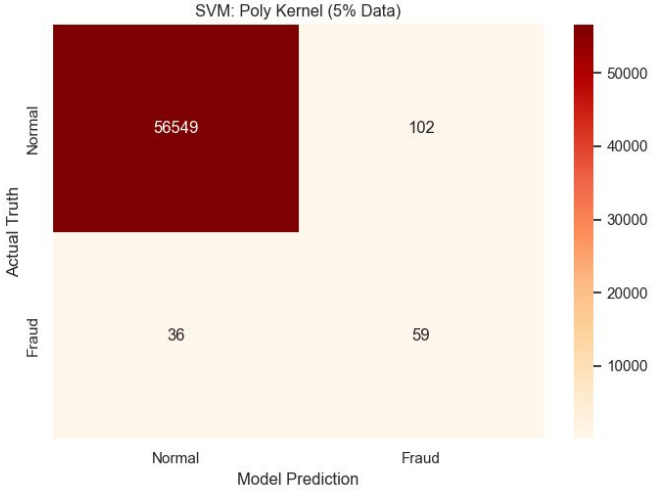
Testing Set: 56,746 rows
Normal: 56,651
Fraud: 95

Best Performing ML Model:
Random Forest









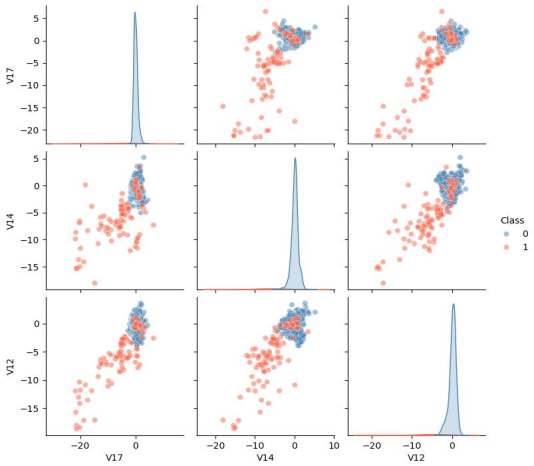
Baseline Observations

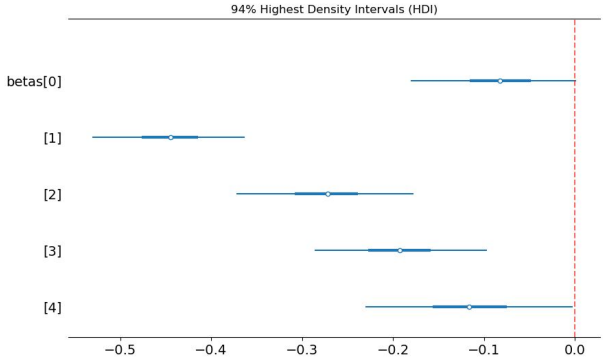
	Model	Catch Rate (Recall %)	Annoyance (Precision %)	F1-Score
0	Logistic Regression (Full Data)	77.89	10.76	0.19
1	Random Forest (Full Data)	72.63	97.18	0.83
2	SVM: RBF Kernel (5% Data)	30.53	34.52	0.32
3	SVM: Poly Kernel (5% Data)	62.11	36.65	0.46

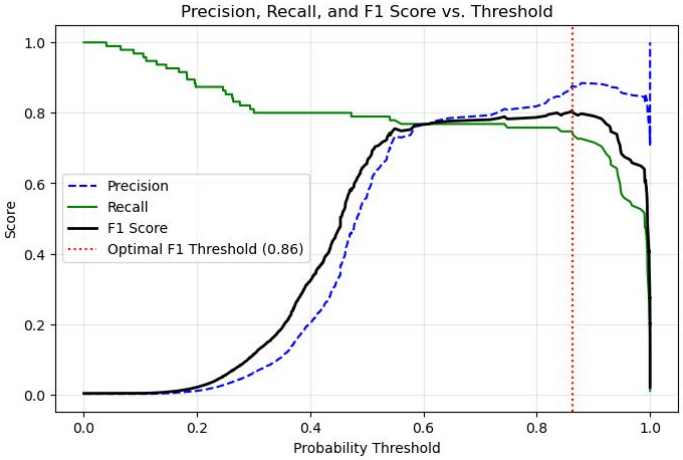
Variables To Select

```
--- The 5 Strongest Signals for Fraud ---
1. V17 (Correlation strength: 0.3135)
2. V14 (Correlation strength: 0.2934)
3. V12 (Correlation strength: 0.2507)
4. V10 (Correlation strength: 0.2070)
5. V16 (Correlation strength: 0.1872)
```

Observations For Objective 1







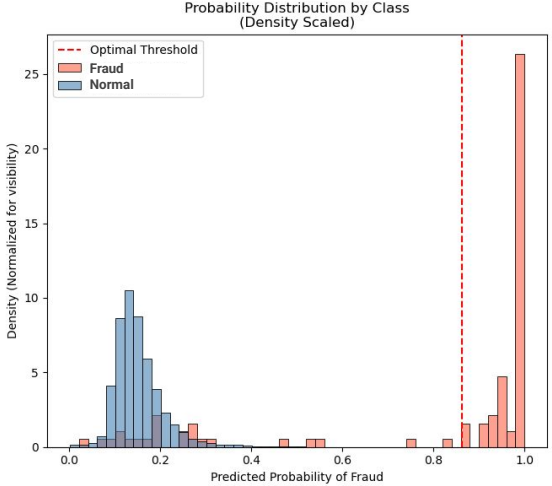
Dataset:

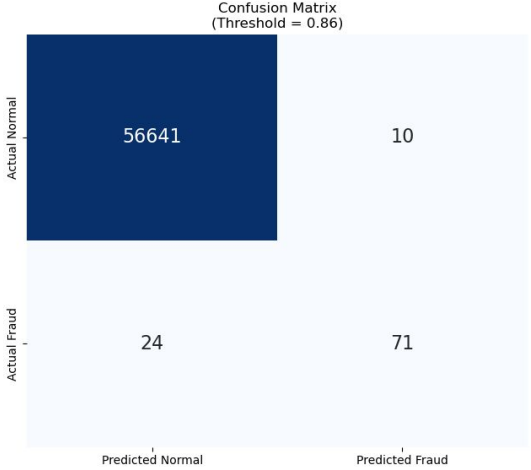
Total rows for Bayesian Sampler: 1890

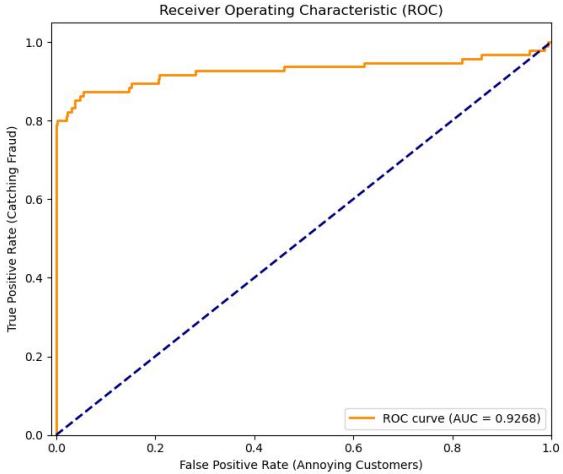
Normal: 1512

Fraud: 378

Results: Objective 1





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Statistically Optimal Threshold: 0.8623
Maximum F1-Score: 0.8068

Confusion Matrix at this Threshold:
True Negatives (Normal passed): 56,641
False Positives (Normal blocked - Annoyed Customers): 10
False Negatives (Fraud missed - Stolen Money): 24
True Positives (Fraud caught!): 71
```

Results: Objective 1 (Cont'd)

```
Total Transactions in Vault: 56,746
Total Actual Frauds: 95
Total Normal Transactions: 56,651
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Q: At what threshold do we capture 100% of frauds?
A: We must lower the threshold to 0.040444 (4.0444%).
   If we do this, we will accidentally block 56,315 normal customers.
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Q: How many normal transactions did the model incorrectly flag with > 90% certainty?
A: 9 highly confident False Positives.
```

Observations For Objective 2

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Total Fraud Attempted in Vault: $14,766.31

STRATEGY A (Static 0.86 Threshold):
  - Friction Cost (Annoyed Customers): $100.00
  - Fraud Lost (Stolen Money): $4,051.53
  - TOTAL FINANCIAL DAMAGE: $4,151.53
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Baseline Cost (Strategy A): $4,151.53
Best Hybrid Cost (Dynamic + Floor): $3,913.24
Optimal Floor Threshold: 0.7227 (72.27%)
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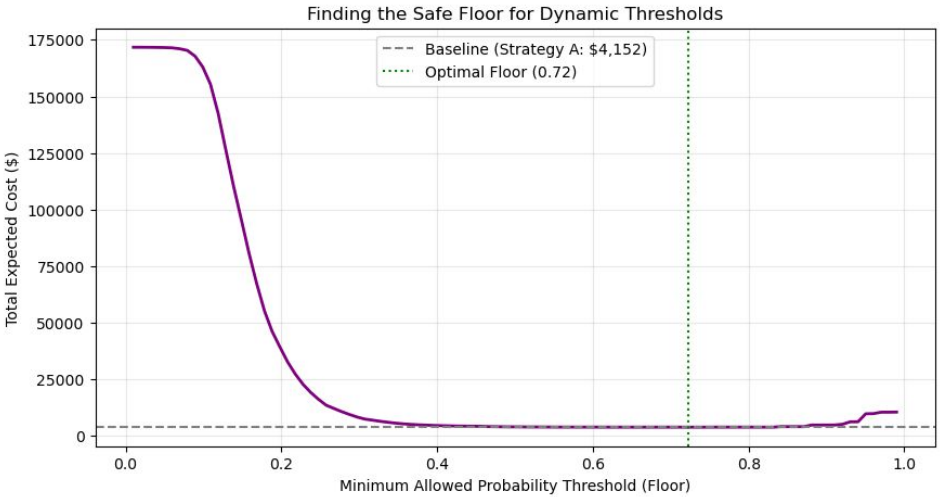
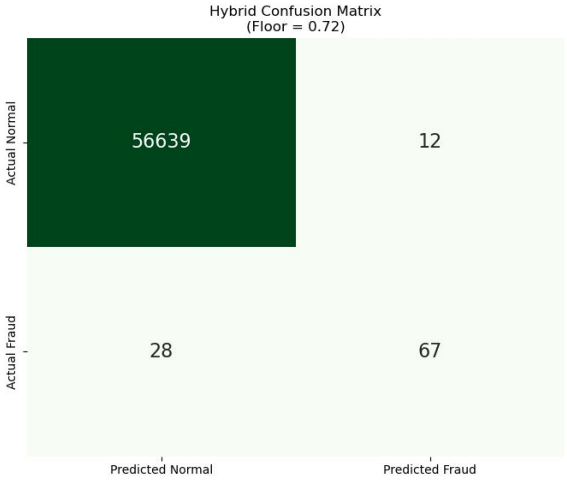
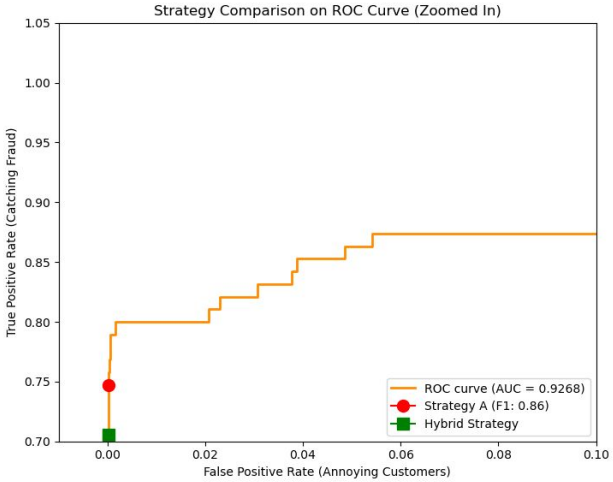


Observations:

Real Net Savings: $238.29

Observations For Objective 2 (Cont'd)

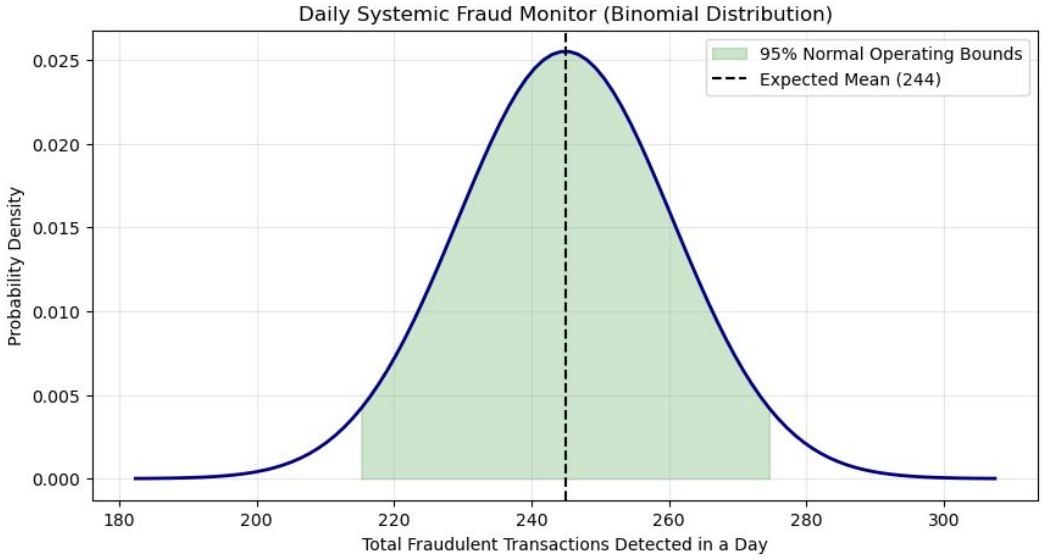




Observations:

Our model performs relatively worse but we are saving more money!

Observations For Objective 3



Observations:

Average Daily Transaction Volume: 142,403
True Base Rate of Fraud: 0.172%
Expected Frauds Per Day: 244
95% Safe Systemic Range: 214 to 275 frauds per day

Limitations & Future Work

Limitations:

1. Computational power and resources.

2. The data was encrypted. Therefore, we were not able to eyeball the best parameters.

Future Work:

1. Model will actually perform better in the hands of financial institutions because of versatile data.

2. Model can be made more efficient if we use RF and MCMC sequentially.

Thank you